

ModellingClub

Software Requirements Specification

Table of Contents

1.Introduction	page 1
2.Feasibility Study	page 2
3.Stakeholders & Actors	page 3
4.Scope & System Overview	page 4
5.Functional Requirements (Use Cases)	page 5
6.Non-Functional Requirements	page 7
7.Domain-Specific Requirements	page 8
8.GUI Wireframes	page 9
9.Process & Behaviour Models	page 10
10.Regulatory & Legal Compliance	page 13
11. Maintenance & Support	page 14
12.Conclusion	page 14

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for ModellingClub, a self-hosted web application that enables hobbyists to register, collaborate on scale-model and drone builds, and showcase their completed creations. The document serves as the primary reference for all development, testing, and maintenance activities.

1.2 Scope

ModellingClub is a lightweight, single-server web application. It is not a commercial platform and carries no revenue model. Key capabilities include member registration, collaborative build tracking, media upload, community voting, and a mandatory drone/UAV flight-log feature required for EU regulatory compliance.

1.3 Intended Audience

- Developer(s) implementing the system
- Self-hoster / system administrator

1.4 Definitions & Acronyms

Acronym / Terminology	Definition
Admin	Administrator; the superuser role
BPMN	Business Process Model and Notation

CRUD	Create, Read, Update, Delete
DMCA	Digital Millennium Copyright Act
EASA	European Union Aviation Safety Agency
FR	Functional Requirement
GDPR	General Data Protection Regulation
IP	Internet Protocol
JWT	JSON Web Token
Nginx	A high-performance web server
ORM	Object-Relational Mapper
PostgreSQL	A powerful, open-source database system
RR	Regulatory Requirement
SLA	Service Level Agreement
SMTP	Simple Mail Transfer Protocol
SysML	Systems Modeling Language
UAV	Unmanned Aerial Vehicle
UUID	Universally Unique Identifier
VPS	Virtual Private Server
WCAG	Web Content Accessibility Guidelines
x86	A family of instruction set architectures

2. Feasibility Study

2.1 Technical Feasibility

The system requires a single Linux server (physical or VM) with Python 3.11+, PostgreSQL 15+, Nginx. All components are open-source and freely available. Docker Compose deployment is recommended for isolation and ease of upgrades.

2.2 Economic Feasibility

Hardware cost: 0zł (re-use existing server/VM) to ~344zł (Raspberry Pi 4 kit). Ongoing cost: electricity (~12.90zł/month), optional domain name (~43zł/year). All software dependencies are free and open-source.

Total first-year cost: < 430zł.

2.3 Legal & Regulatory Feasibility

The platform stores personal data (email, username) and must comply with GDPR. Drone builds require flight log records per EU Regulation 2019/945 and 2021/664. Content moderation is required to prevent posting of illegal modifications (e.g. beyond Open Category A3 limits).

These requirements are tractable at hobby scale.

2.4 Schedule Feasibility

Phase	Duration	Deliverable
Setup & Auth	2 week	User registration, login, JWT sessions
Build Management	2 weeks	CRUD builds, media upload, status workflow
Collaboration	2 week	Invitations, co-author roles
Drone / Flight Logs	2 week	Flight log form, regulatory checks
Showcase & Community	2 week	Public gallery, voting, comments
Admin Panel & Moderation	2 week	User management, content review
Testing & Hardening	1 week	Integration tests, security review

3. Stakeholders & Actors

Actor	Role	Access Level
Guest	Unauthenticated visitor; browses public gallery	Read-only

Member	Registered user; creates and collaborates on builds	Full user
Admin	Platform operator; manages users and content	Superuser
Email Service	SMTP relay (local Postfix or Mailpit in dev)	System
File System	Local storage for media uploads	System

4. Scope & System Overview

ModellingClub is a web-based platform for scale-model and drone hobbyists. Members register, create & build projects, upload photos and CAD files, collaborate with other members, and publish their finished work in a public gallery. For drone/UAV projects the system enforces flight-log entry as required by EUlaw.

4.1 In Scope

- User registration, authentication (SHA + JWT sessions)
- Build project CRUD with status workflow (Draft → Review → Published / Rejected / Archived)
- Media upload (JPEG, PNG, PDF; max 20 MB per file; max 50 files per build)
- Collaboration invitations and co-author roles
- Drone / UAV flight log (mandatory for UAV-type builds)
- Public gallery with search, filtering, and community voting
- Comment threads on builds
- Admin panel: user management, content moderation queue
- Email notifications (registration, review outcome, invitations)
- GDPR data-export and account-deletion flow

4.2 Out of Scope

- Native mobile applications (responsive web only)
- Real-time video streaming
- E-commerce / payments
- Integration with external social networks
- Official CAA/EASA certification portal integration

5. Functional Requirements

5.1 Use Case Diagram

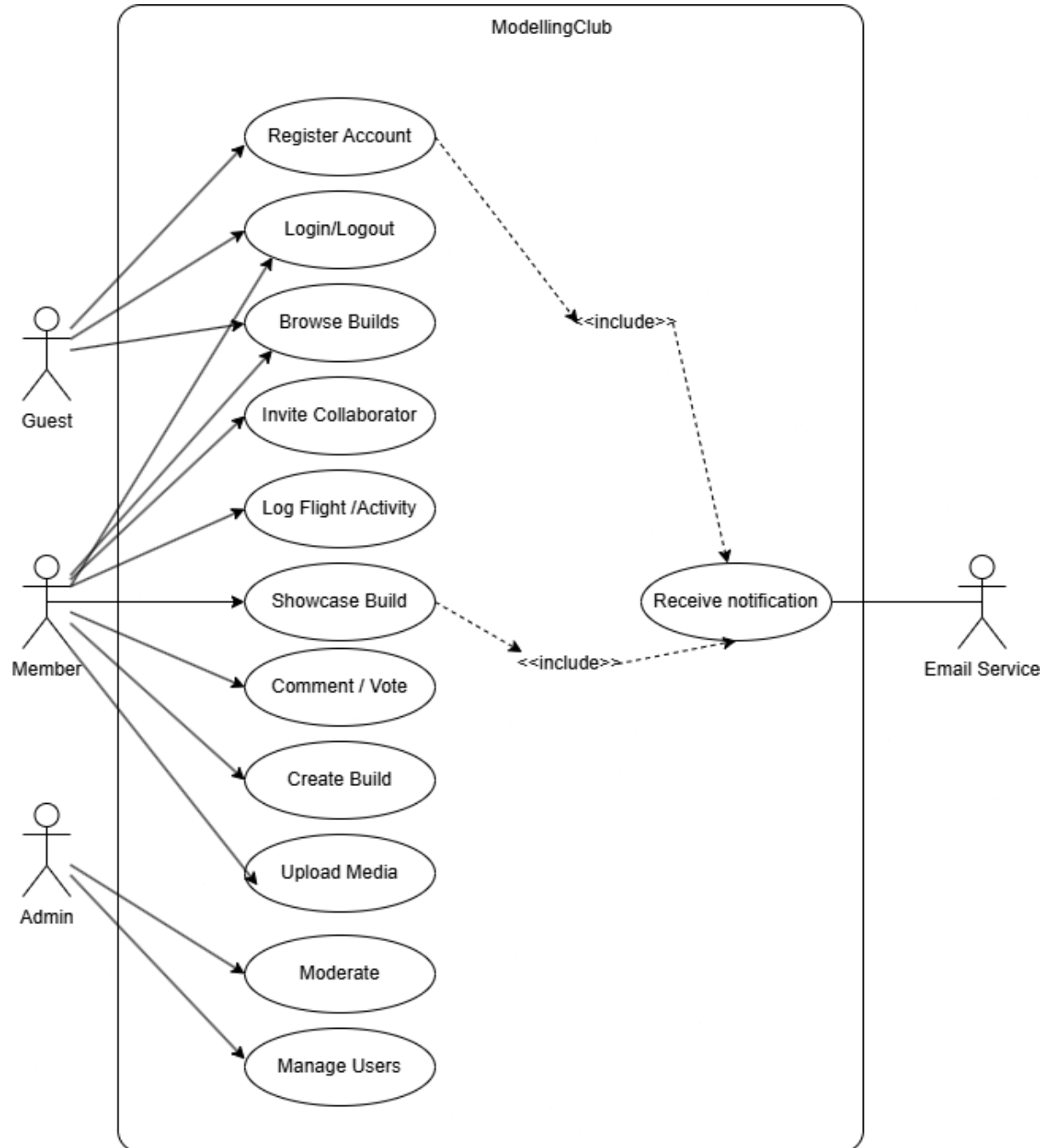


Figure 1: ModellingClub Use Case Diagram. Overview of actor interactions with the system.

5.2 Use Case Descriptions

ID	Name	Description	Constraints / Notes
UC-01	Register Account	Guest submits username, email, password. System validates, stores hashed password, sends confirmation email.	Email not yet verified; duplicate username/email rejected.
UC-02	Login / Logout	Members submit credentials. System verifies SHA hash, issues JWT session cookie. Logout invalidates the session.	Invalid credentials return 401; brute-force lockout after 5 attempts/15 min.
UC-03	Create Build	Member fills title, type (Fixed-Wing / Drone / Helicopter / Boat / Car / Other), description, scale, dimensions.	Title ≤120 chars; type is mandatory.
UC-04	Upload Media	Member uploads files to an existing Build. System validates type and size, stored on filesystem.	Max 20 MB/file; allowed: JPEG, PNG, GIF, PDF, STL; max 50 files/build.
UC-05	Invite Collaborator	Build owner emails invitation link to another Member. Invitee accepts; grants system co-author role.	Link expires in 48 h; invitee must be a registered Member.
UC-06	Submit for Review	Member submits a Draft build. System changes status to Pending Review and notices Admin.	Must have title, type, ≥1 photo, and (for UAV) ≥1 flight log entry.
UC-07	Approve/Reject	Admin reviews build. Approves (status -> Published) or rejects with mandatory reason (status -> Rejected).	Admin only; reason text ≤500 chars.
UC-08	Log Flight Activity	Member records a UAV flight: date, location, duration, max altitude, drone identifier.	Mandatory for Drone-type builds; altitude check per EU 2019/945.
UC-09	Browse & Search	Guest or Member searches/filters published builds by type, keyword, author.	Full-text search via PostgreSQL tsvector.
UC-10	Vote & Comment	Members upvotes a build and posts comments.	Self-voting disallowed; comments ≤1000 chars.
UC-11	GDPR Export/Delete	Members request data export (JSON) or account deletion (anonymization).	Export within 24 h; deletion is irreversible.

6. Non-Functional Requirements

6.1 SysML Requirements Overview

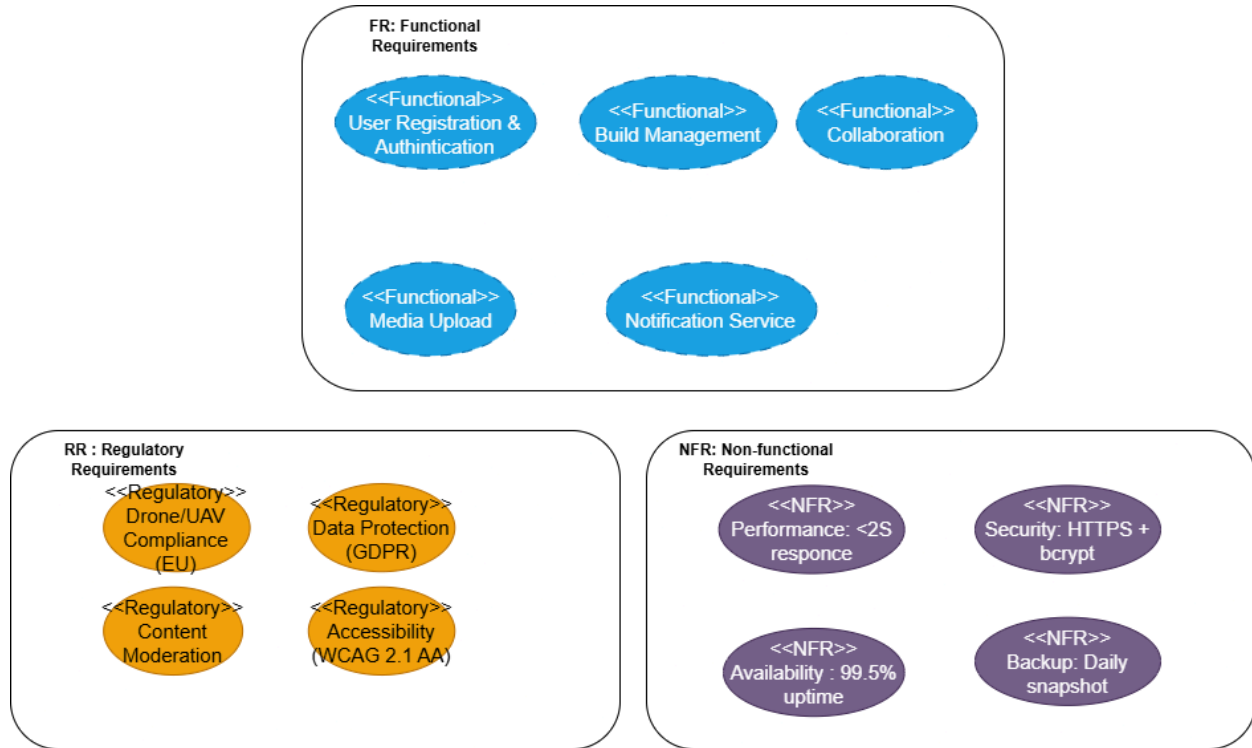


Figure 2: SysML Requirements Map. Visual categorization of FR, RR, and NFRs.

6.2 Non-Functional Requirements Table

ID	Category	Requirement	Measure/Target
NFR-001	Performance	Page load time (server-side rendered)	≤2 s at P95 under 50 concurrent users
NFR-002	Performance	Media upload throughput	≥5 MB/s on local network
NFR-003	Scalability	Max concurrent users (single server)	100 (Raspberry Pi 4) / 500 (x86 VPS)
NFR-004	Security	All HTTP traffic encrypted	TLS 1.2+ via Nginx
NFR-005	Security	Password storage	SHA, cost factor ≥12
NFR-006	Security	Session management	HttpOnly + SameSite=Lax cookies; 7-day expiry
NFR-007	Security	File upload validation	MIME type + magic-bytes check; no executables
NFR-008	Availability	Planned uptime	99.5% (≤44 h downtime/year)
NFR-009	Reliability	Database backup	Daily pg_dump; 30-day retention
NFR-010	Usability	Accessibility	WCAG 2.1 Level AA
NFR-011	Usability	Mobile responsiveness	Layout functional at 320 px viewport width

7. Domain-Specific Requirements

7.1 Build Types & Attributes

Build Type	Attributes Mandators	Optional Attributes
Fixed-Wing	Title, wingspan_mm, scale, engine_type	fuselage_length_mm, motor_kv
Drone / UAV	Title, frame_size_mm, num_motors, flight_controller, max_takeoff_weight_g	camera_type, fpv_freq_mhz
Helicopter	Title, rotor_diameter_mm, scale	collective_pitch
Boat	Title, hull_length_mm, motor_type	displacement_g
Tsar / Ground	Title, wheelbase_mm, scale	gearbox_type
Scale Model	Title, scale (eg 1:72), subject	material, manufacturer_kit
Other	Title, description	—

7.2 Flight Log Fields & Validation Rules

Field	Type	Validation Rule
flight_date	DATE	Not in future; not before build creation date
location_name	TEXT	1–200 characters; free text
duration_min	INT	1–600 (max 10 h per log entry)
max_altitude_m	DECIMAL	0–120 m (Open Category limit); warning > 50 m
drone_identifier	TEXT	Regex: <code>^[AZ]{2}-[AZ0-9]{6,12}\$</code> (EASA pattern)
conditions	ENUM	SUNNY, CLOUDY, WINDY, RAIN — if RAIN, advisory show
notebook	TEXT	0–1000 characters; optional

7.3 Voting & Scoring Formula

Build visibility score used for gallery ranking is computed as:

$$\text{score} = \text{upvotes} - \text{downvotes} + \log_{10}(\max(1, \text{comments})) * 2 + (\text{days_since_published} < 7 ? 5 : 0)$$

This is a simplified variant of the Wilson score interval, favouring recent and engaged builds without requiring statistical sampling.

8. GUI Wireframes

8.1 Dashboard / Home Page

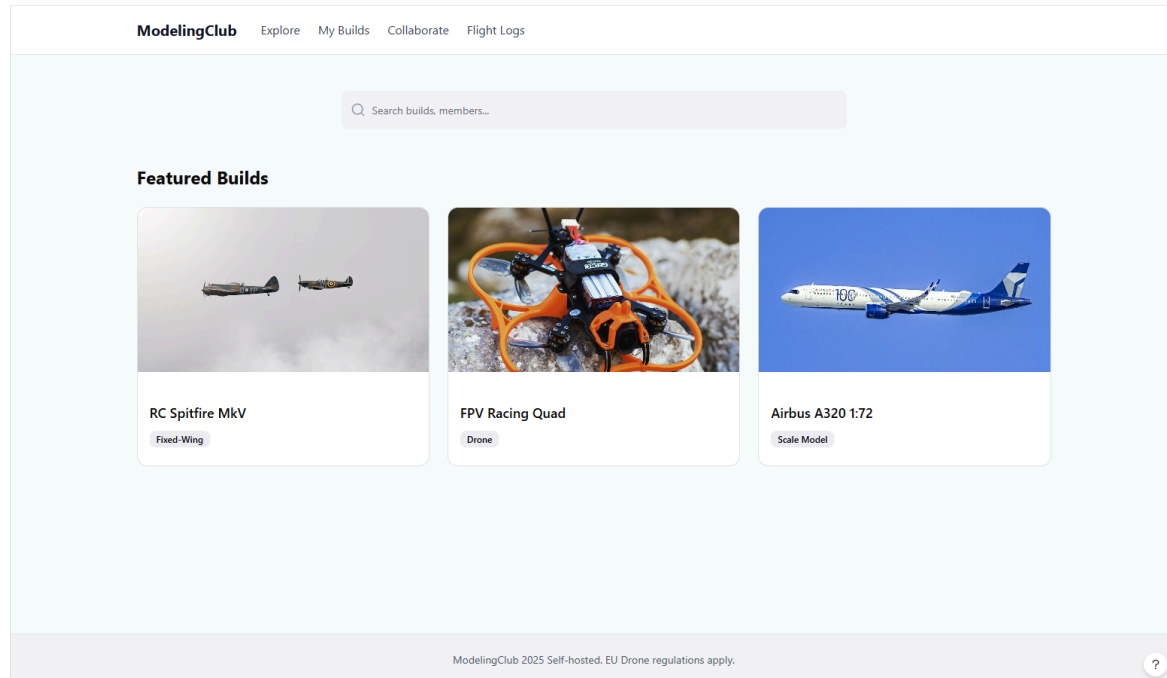
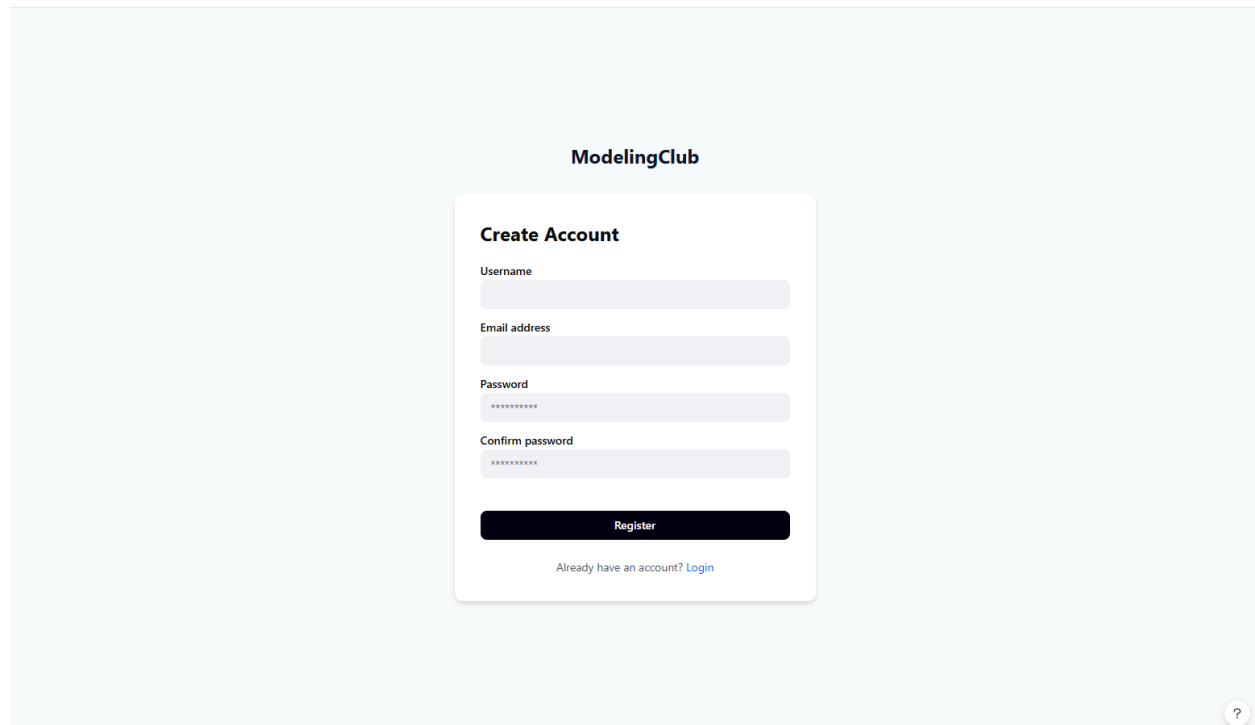


Figure 3: Dashboard/ Home Page Wireframe. The primary user interface for exploring builds

8.2 Registration & New Build Forms



ModelingClub Explore My Builds Collaborate Flight Logs [+ New Build](#)

New Build

Build Title *

Type *
Select type

Description

Scale (e.g. 1:72)
1:72

Wingspan / Size (mm)
1000

Photos / CAD files
Drop files or [click to browse](#)

[Save Draft](#) [Submit for Review](#)

Figure 4&5: Registration & New Build. The primary user interface for registration and submitting new build

9. Process & Behaviour Models

9.1 Sequence: User Registration

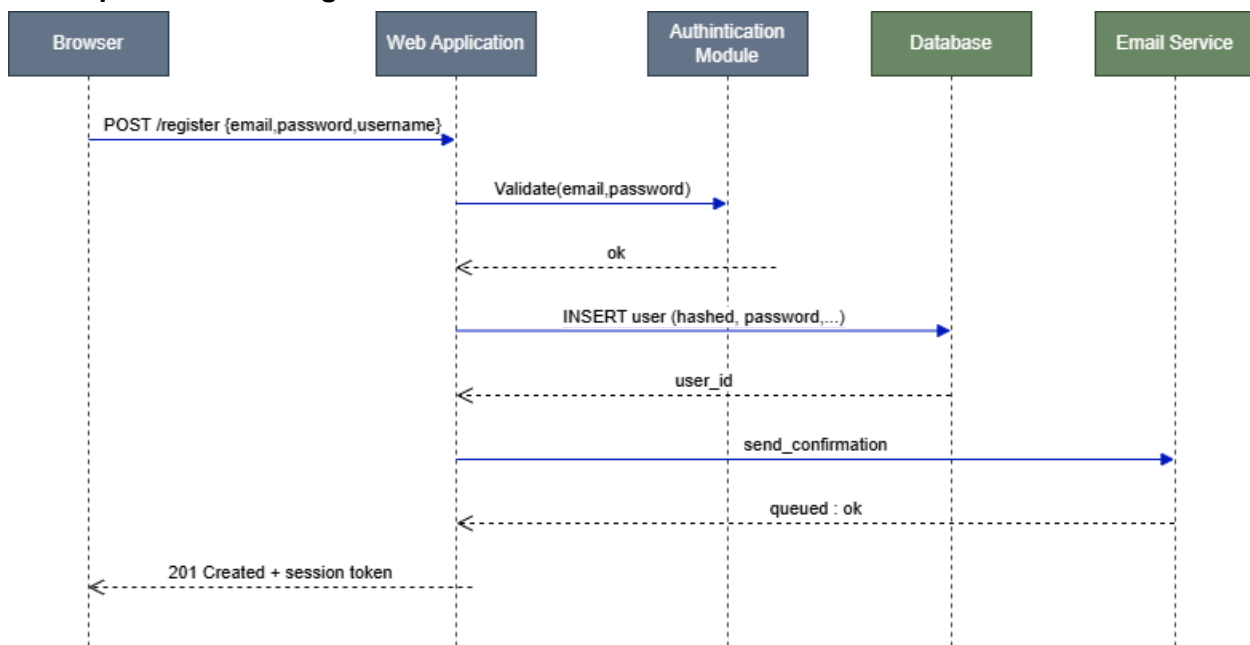


Figure 6: User Registration Sequence. Logic flow from browser to database and email

9.2 Sequence: Create & Submit Build

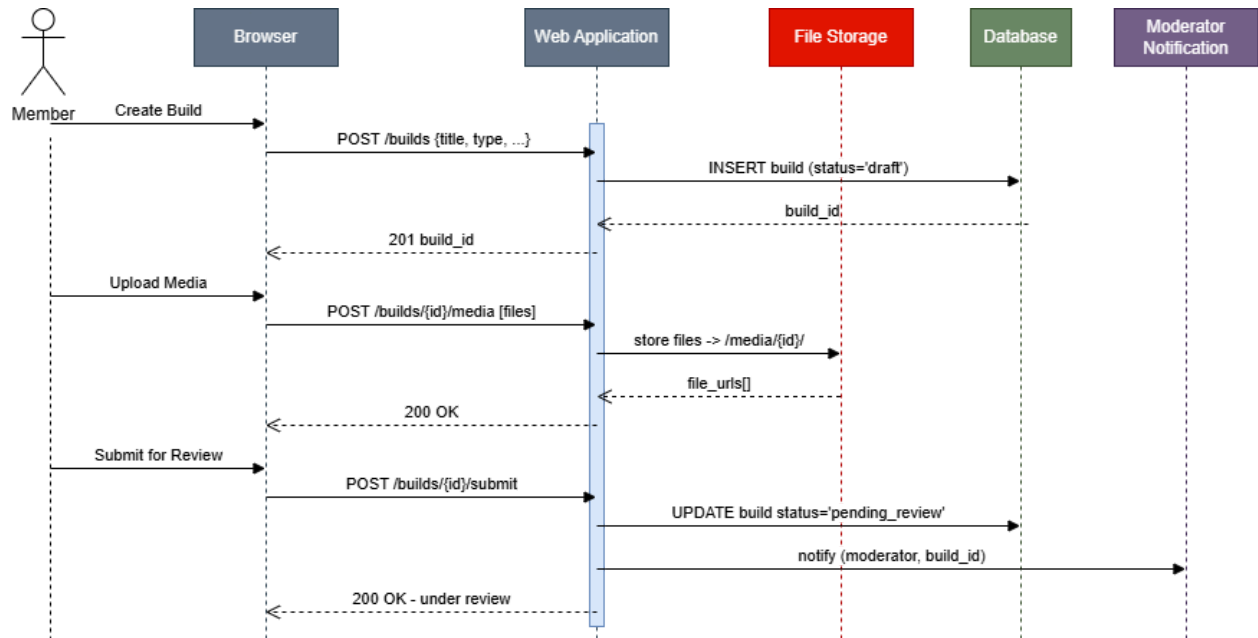


Figure 7: Build Workflow Sequence. Path from draft creation to admin review

9.3 State Machine: Build Lifecycle

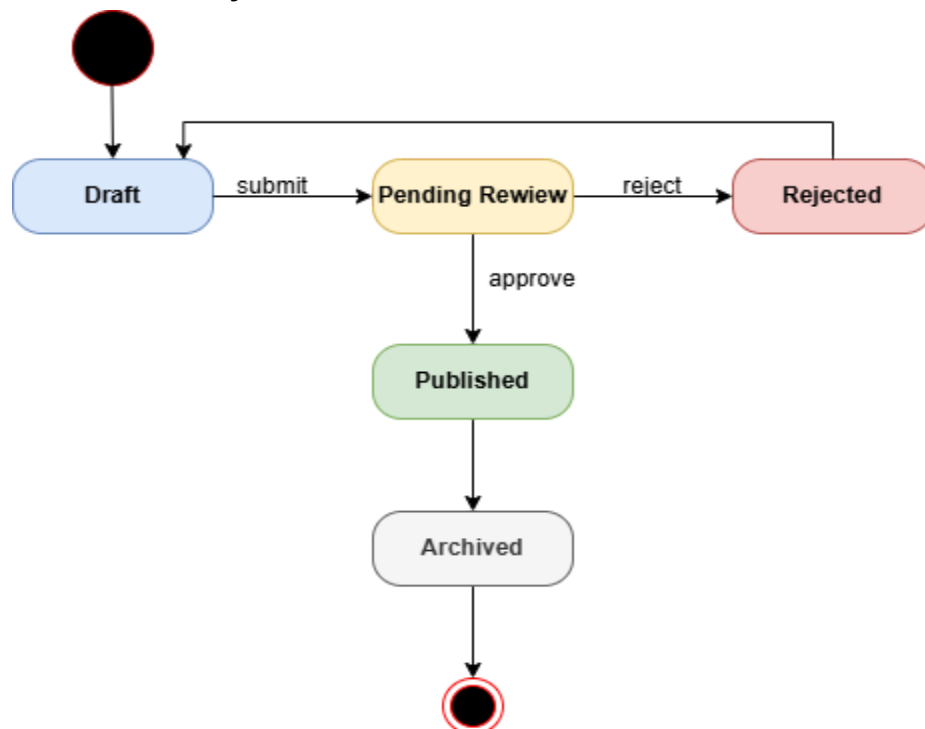


Figure 8: Build State Machine. The lifecycle and status transitions of a project

9.4 BPMN: Drone Flight Log Submission

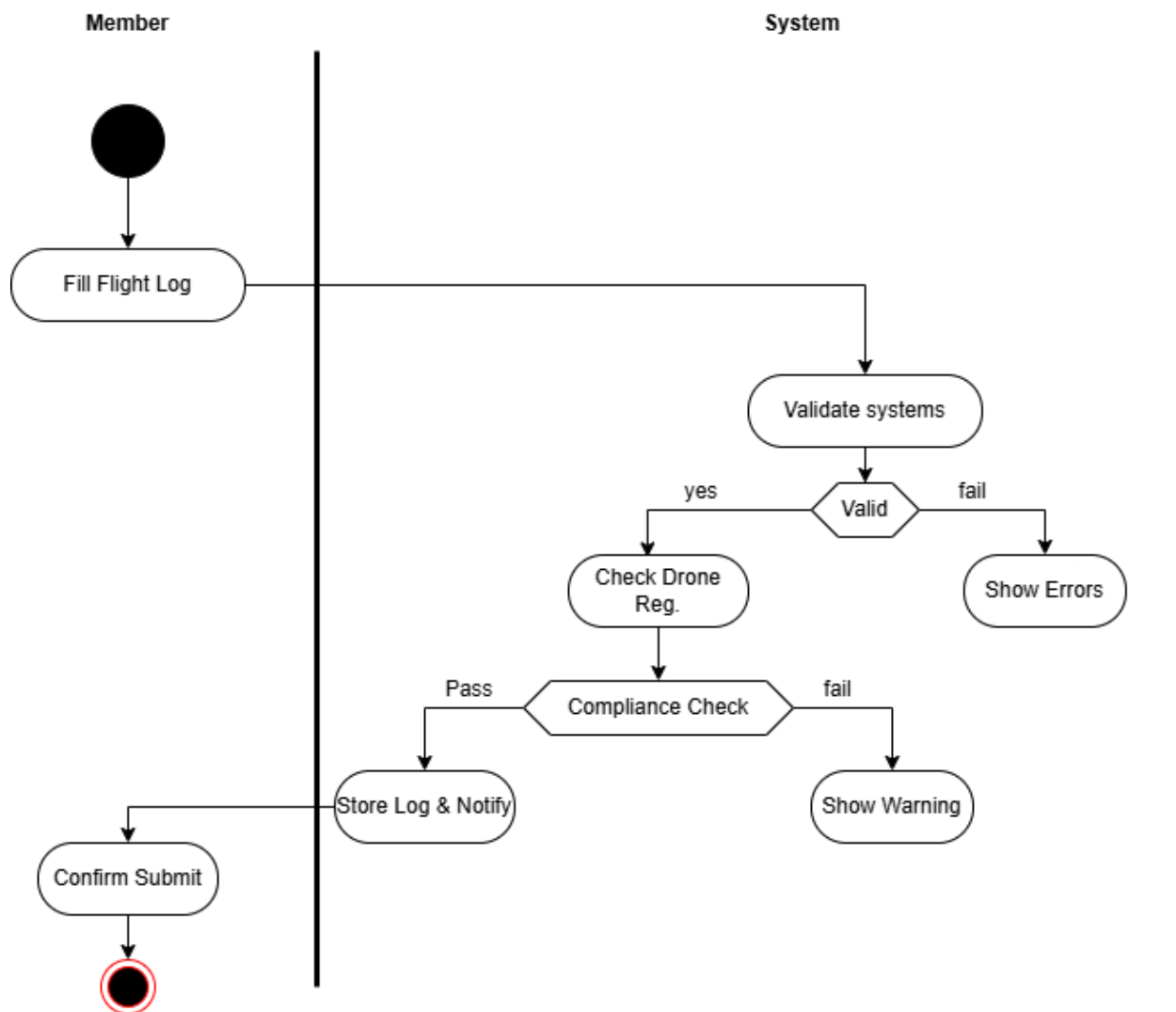


Figure 9: Flight Log BPMN. Logical validation process for drone flight data

10. Regulatory & Legal Compliance

10.1 EU Drone Regulations (EU 2019/945 & 2021/664)

Any build categorised as a UAV/Drone must comply with EU UAS regulations. The system enforces the following rules:

- Maximum altitude logged must not exceed 120 m AGL (Open Category A1–A3 limit). The system warns and blocks submission if any log entry exceeds this value.
- Drone identifier field is mandatory and must match EASA UAS operator registration format pattern.
- The platform displays a regulatory information banner on all drone build pages linking to the current EASA rules.
- The system does not replace or substitute official registration with national CAA/EASA portals.

10.2 GDPR (EU 2016/679)

Article	Implementation
Art. 6 – Lawful basis	Registration collects only email + username; consent recorded with timestamp
Art. 13/14 – Information	Privacy notice displayed at registration and in footer
Art. 17 – Right to erasure	Account deletion anonymises personal data; builds attributed to '[deleted]'
Art. 20 – Portability	Data export (JSON) available on request via account settings
Art. 25 – Privacy by design	Minimal data collection; no tracking cookies; no third-party analytics
Art. 32 – Security	Encrypted storage, SHA passwords, HTTPS-only

10.3 Content & Intellectual Property

- Members retain copyright on uploaded media; they grant ModellingClub a non-exclusive licence to display content on the platform.
- Admin moderation queue reviews all submissions before publication.
- Uploads of illegal weapon modifications, jailbreaks, or commercially licensed CAD without authorisation is prohibited and enforced by admin review.
- DMCA-equivalent take-down procedure: admin can unpublish and archive any build.

11. Maintenance & Support

11.2 Upgrade Procedure

1. Pull latest source from repository (git pull).
2. Review CHANGELOG.md for breaking changes.
3. Run database migrations: flask db upgrade.
4. Smoke test: curl https://localhost/health → 200.

11.2 Helpdesk & Issue Reporting

ModellingClub is self-hosted and community-supported. Issues are tracked via the project's GitHub Issues page. The admin user receives email alerts for application ERROR-level log events. No commercial SLA is offered.

12. Conclusion

The ModellingClub platform is designed to bridge the gap between technical precision and community collaboration. By providing a secure, self-hosted environment, this system empowers hobbyists to document their builds, share technical knowledge, and maintain regulatory compliance with ease.

ModellingClub is more than just a piece of software; it is a commitment to the hobby. By adhering to EASA standards and prioritizing data sovereignty, we ensure that as the technology of our builds evolves, the tools we use to document and celebrate them remain robust, compliant, and sustainable.